

Rectangles & Squares — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

What type of quadrilateral has four right angles and four equal sides?

- (a) Rectangle (b) Square (c) Rhombus (d) Trapezium

Q2 • MCQ**[1]**

The diagonals of a rectangle are always:

- (a) equal and bisect each other (b) perpendicular to each other (c) unequal (d) parallel

Q3 • FILL IN THE BLANK**[1]**

The diagonals of a rectangle are $(2x + 1)$ and $(3x - 1)$. The value of x is:

Q4 • MCQ**[1]**

In which figure do the diagonals meet at right angles AND are equal in length?

- (a) Rectangle (b) Parallelogram (c) Square (d) Trapezium

Q5 • TRUE / FALSE**[1]**

Every square is a rectangle.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): The diagonals of a square are equal.

Reason (R): A square is a special kind of rectangle.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Find the length of the diagonal of a square whose side is 7 cm. (Use $\sqrt{2} \approx 1.41$.)

Q8

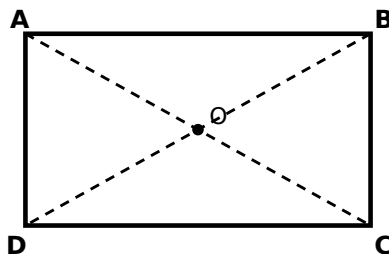
[2]

A rectangle has sides 9 cm and 12 cm. Find the length of its diagonal.

Q9 · FIGURE

[2]

In rectangle ABCD the diagonals meet at O. If $OA = 5$ cm, write the lengths of OC, OB, OD and the full diagonal AC. Justify briefly.



Long answer (3 marks each)

Q10

[3]

The diagonals of a rectangle are $(5x - 3)$ cm and $(3x + 5)$ cm. Find x and the length of each diagonal.

Q11

[3]

A square has a diagonal of length 6 cm. Without a protractor, explain how you would construct it, and state the angle at which the diagonals meet.

Q12

[3]

State, with reasons, which of these are always true for a rectangle: (i) diagonals equal, (ii) diagonals perpendicular, (iii) all sides equal, (iv) diagonals bisect each other.

Total: 21 marks · Check your work against the answer key.